

Confidence Intervals

LLE – Mathematics and Statistics Skills

1. Let's say that a researcher is interested in the mean length of time people spend using a particular product each year. They collect a sample from the population of people who have the product and calculate the mean length to be 6.4 days.

- (a) With the information given, what would be the researcher's estimate of the population mean number of days people are using the product?

Solution

The best estimate for the population mean would be the sample mean: 6.4 days

- (b) Using other statistics collected from the sample, the researcher also calculates a 95% confidence interval for the population mean. It is calculated as (4.6, 8.2).

What is the interpretation of this confidence interval?

- ☐ We are 5% confident that the true mean is between 4.6 and 8.2 days
- ☒ **We are 95% confident that the true mean is between 4.6 and 8.2 days**
- ☐ We are 95% confident that the true mean is outside of 4.6 and 8.2 days
- ☐ We have no idea how many days people use the product for for on average

2. The margin of error in a confidence interval is determined by the level of confidence, the sample size, and the standard deviation. When the population standard deviation is known you can use the formula, for a 95% confidence interval:

$$\bar{x} \pm 1.96 \times \frac{\sigma}{\sqrt{n}}$$

Where $\bar{x}$ is the sample mean, σ is the population standard deviation, n is the sample size.

- (a) Calculate the 95% confidence interval if sample mean = 40, population standard deviation = 10, sample size = 100

Solution

The margin of error is $1.96 \times \frac{10}{\sqrt{100}} = 1.96$
so $40 - 1.96 = 38.04$ and $40 + 1.96 = 41.96$

- (b) Calculate the 95% confidence interval if sample mean = 100, population standard deviation = 15, sample size = 25

Solution

The margin of error is $1.96 \times \frac{15}{\sqrt{25}} = 5.88$
so $100 - 5.88 = 94.12$ and $100 + 5.88 = 105.88$

- (c) Calculate the 95% confidence interval if sample mean = 100, population standard deviation = 15, sample size = 225

Solution

The margin of error is $1.96 \times \frac{15}{\sqrt{225}} = 1.96$
so $100 - 1.96 = 98.04$ and $100 + 1.96 = 101.96$

- (d) Calculate the 95% confidence interval if sample mean = 100, population standard deviation = 75, sample size = 225

Solution

The margin of error is $1.96 \times \frac{75}{\sqrt{225}} = 9.2$
so $100 - 9.2 = 90.8$ and $100 + 9.2 = 109.2$

3. Other size confidence intervals can be calculated by changing the 1.96 part of the formula. The correct value can be obtained from normal tables.

- (a) What is the critical value for a 99% confidence interval of the mean? Give answer to 3 d.p. and positive.

Solution

Using tables gives 2.576

- (b) Given a sample mean of 100, population standard deviation of 30, and sample size of 225, what is the 99% confidence interval of the mean?

Solution

$100 \pm 2.576 \frac{30}{\sqrt{225}}$
gives (94.85, 105.15)

4. If you do not have the population standard deviation when calculating a confidence interval for the mean, you can make use of the standard deviation of the sample. When doing this you will need to make use of the t-distribution, rather than the normal distribution.

From a sample of size 20, a sample mean of 156 and a sample standard deviation of 12.2 is found.

- (a) What would be the critical t value for a 95% confidence interval, correct to 3 d.p.?

Solution

Use the calculator or tables, with one-sided 0.025, and degrees of freedom of 19, gives the value of 2.093

- (b) What would be the 95% confidence interval for the mean?

Solution

$156 \pm 2.093 \times \frac{12.2}{\sqrt{20}}$
(150.29, 161.71)

5. From a sample of size 12, a sample mean of 100 and a sample standard deviation of 15 is found.

What is the 90% confidence interval for the mean, correct to 2 d.p.?

Solution

Critical t value is 1.796,

$$\text{so } 100 - 1.796 \times \frac{15}{\sqrt{12}} = 92.22$$

$$100 + 1.796 \times \frac{15}{\sqrt{12}} = 107.78$$

6. (a) Complete the table of confidence intervals for the sample statistics, correct to 1 decimal place.

Size	Mean	Std. Dev.	95% Conf. Int.	99% Conf. Int.
15	50	4	47.8, 52.2	46.9, 53.1
15	50	15	41.1, 58.3	38.5, 61.5
10	50	15	39.3, 60.7	34.6, 65.4

- (b) What do you notice to the width (difference between upper and lower bounds) of the confidence interval as the confidence percent increases?

☐ As the confidence percent increases, the width of the interval decreases

☒ **As the confidence percent increases, the width of the interval increases**

- (c) What do you notice to the width (difference between upper and lower bounds) of the confidence interval as the sample size increases?

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☐ As the sample size increases, the width of the interval increases

7. A class from the Environmental Science School collects 20 samples of water from a lake. The mean pH value of the 20 samples is 8.2. The local council collects 100 samples from the lake and gets a mean pH of 8.

It is known that the standard deviation of the pH values for the lake is 0.6.

- (a) Calculate 95% confidence intervals for the mean pH from the samples from the environmental science class

Solution

Use normal tables, due to known standard deviation, with $z = 1.96$
(7.94, 8.46)

- (b) Will the local council have a wider 95% confidence interval than the science class?

Solution

The local council will have a smaller gap between the lower and upper bound, due to a larger sample size.

8. A random sample of 15 callers to the “Pecs Bar & Gym” support line showed that they were on hold for the following number of minutes:

19, 28, 34, 10, 27, 31, 25, 37, 54, 27, 54, 8, 17, 24, 25

- (a) What is the mean waiting time for the sample?

Solution

$$\bar{x} = \frac{420}{15} = 28$$

- (b) What is the standard deviation of waiting time for the sample, correct to 2 d.p.?

Solution

Use an online calculator or the formula $s = \frac{\sum(x-\bar{x})^2}{n-1} = 13.20$

- (c) Given that the confidence interval is written as Mean $\pm$ Error, what is the value of the Error, to 2 d.p.?

Solution

The critical t-value is 2.145, so Error = $2.145 \times \frac{13.2}{\sqrt{15}} = 7.31$

9. A random sample of measurements is taken from a normal distribution with an unknown mean and a known population standard deviation of 15.

- (a) What is the smallest sample size so that the width of a 95% confidence interval is smaller than 6 units?

Solution

Margin of error needs to be 3

$$1.96 \times \frac{15}{\sqrt{n}} = 3$$

$$\sqrt{n} = \frac{1.96 \times 15}{3}$$

$$n = 96.04$$

Needs to be over 96.04, so 97.

- (b) What is the smallest sample size so that the width of a 95% confidence interval is smaller than 3 units?

Solution

Margin of error needs to be 1.5

$$1.96 \times \frac{15}{\sqrt{n}} = 1.5$$

$$\sqrt{n} = \frac{1.96 \times 15}{1.5}$$

$$n = 384.16$$

Needs to be over 384.16, so 385.